

Ch1. Review

Model: $\mathbf{Y} = \mathbf{X}\beta + \mathbf{u}$

Estimator: $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$

Partitioned Regression: $\hat{\beta}_1 = \left(\sum \tilde{X}_{i1}\tilde{X}'_{i1}\right)^{-1} \left(\sum \tilde{X}_{i1}\tilde{Y}_{i1}\right)$, wavytilde means residual on X_2

Ch2. Endogeneity

Endogenous: $\text{Cov}(X_i, u_i) \neq 0$

Sources: omitted variable(OVB), add regressor/panel data/IV/randomized controlled; wrong model, change model; measurement error(biased toward 0), better data/model error/IV; simultaneity, IV

Ch3. IV&2SLS

IV Def: Z_i s.t. $\text{Cov}(Z_i, X_i) \neq 0, \text{Cov}(Z_i, u_i) = 0$ $\hat{\beta}_{IV} = \frac{\sum (Y_i - \bar{Y})(Z_i - \bar{Z})}{\sum (X_i - \bar{X})(Z_i - \bar{Z})}$

Biased(generally biased), consistent, asymptotic normal $N\left(0, \frac{\sigma_v^2}{n\text{cov}(Z, X)^2}\right)$

$\sigma_v = \text{Std}(u_i(Z_i - \mu_Z))$. Under homoskedasticity $\text{Var}(\hat{\beta}_{IV}) = \text{Var}(\hat{\beta}_{OLS}) \cdot \rho_{X,Z}^{-2}$

2SLS: Regress X s on Z s and exogenous W s, then regress Y on fitted X s and original W s. Same as IV.

Must include all W to avoid endogeneity in the second stage.

Structural model: model with endogenous X .

With k X , r W , m Z , when $m=k$ exactly identified, m higher over-identified, m smaller under-identified

IV relevance: get F-stat of Z s in the first regression, compare to 10. If weak IV, not consistent/normal.

IV exogeneity: only if over-identified. Regress residual on m Z s and r W s, $J=mF(F \text{ of } Z\text{s})$ follows $\chi^2(m-k)$

Hausman test: checks if IV and OLS estimates are different.

Endogeneity test: use first stage residual v , regress Y to X, W, v , check if v significant.

Ch4. Binary

LPM: $E(Y|X) = \beta_1 X + \beta_0$. Simple, similar statistical inference to OLS, but not linear/improper value

Probit/Logit: $P(Y = 1|X) = G(\beta_1 X + \beta_0)$. Normal/Logistic CDF.

Non-linear LS: consistent, asymptotically normal, inefficient; MLE, consistent, asymptotically normal, efficient

$\widehat{PEA}_j = g(\bar{x}\hat{\beta})\hat{\beta}_j, \widehat{APE}_j = n^{-1} \sum g(x_i\hat{\beta})\hat{\beta}_j$

Pseudo R-squared $\hat{R}^2 = 1 - \frac{L_{ur}}{L_0}$, compare log-likelihood

LR test: $2(L_{ur} - L_0) \sim \chi^2(q)$

Ch5. Panel Data

DID, DDD estimator to eliminate trend. Panel data, multiple observations for same entity same time.

First-difference estimator: eliminate fixed effect. Fixed effect (within) estimator, demeaned within time.

Need strict exogeneity, $df=nT-n-k$

RE estimator: $(Y - \lambda\bar{Y}) = \sum \beta_i(X_i - \lambda\bar{X}_i) + (a_i + u_{it} - \lambda\bar{a}_i - \lambda\bar{u}_i), \lambda = 1 - \sqrt{\frac{\sigma_u^2}{\sigma_u^2 + T\sigma_a^2}}$

Consistent and asymptotically efficient under RE assumptions. Less bias, removes some serial corr

Hausman's Test: Apply RE&FE, test difference in estimates. Test for serial corr: xtserial

Need clustered SE. $\text{Se}(\bar{Y}) = \sqrt{\frac{s_{\bar{Y}}^2}{n}}, s_{\bar{Y}}^2 = \frac{\sum (Y_i - \bar{Y})^2}{n-1}$

Hausman-Taylor approach: $Y_{it} = X_{1,it}\beta_1 + X_{2,it}\beta_2 + Z_{1,i}\gamma_1 + Z_{2,i}\gamma_2 + a_i + u_{it}$, a_i allowed to be correlated with X_2 and Z_2 . Use IV: $(\ddot{X}_1, \ddot{X}_2, \bar{X}_1, Z_1)$

Ch6. Time Series

Autocovariance, autocorrelation. Distributed lag models: include lag terms in the model. Coef of lag-0 is called impact propensity, of lag-n called dynamic multipliers, all coef added is called long-run propensity. Add trend in model = eliminate trend in Y and X (use regressed residual).

Seasonality: add seasonal dummies.

Assumptions: 1. Linear 2. Full rank 3. Strict exogeneity 4. Homoskedasticity 5. No serial corr 6. Normal

Ch7. OLS in Time Series Data

Stationary. Weakly dependent: $\text{Corr}(X_t, X_{t+h}) \rightarrow 0$

Assumptions: 1. Linear, and Variables stationary and weakly dependent 3. Contemporaneously exogenous 245 Same.

Use HAC standard error if TS5 violated. Order of integration: difference needed to be weak independent Both unit root and trend can be eliminated by differencing; former called stochastic trend, latter drift.

Ch8. Serial Correlation and Heteroskedasticity in Time Series

OLS not BLUE, SE-related not useful; goodness of fit still works

Test for serial correlation: Test significance of residual to lagged residual. Need exogenous.

DW Test: $DW = \frac{\sum (\hat{u}_t - \hat{u}_{t-1})^2}{\sum \hat{u}_t^2}$, reject if $DW > d_L$, fail to reject if $DW > d_U$ Need exogenous.

Extend to AR(q): $\hat{u}_t = \alpha_0 + \alpha_1 x_{t1} + \dots + \alpha_k x_{tk} + \rho_1 \hat{u}_{t-1} + \dots + \rho_q \hat{u}_{t-q}$, $H_0 : \forall \rho_i = 0$

Q test: $Q_n = n \sum \hat{\rho}_j^2 \rightarrow_d \chi^2(p)$ Modified: $Q_n = n(n+2) \sum \frac{\hat{\rho}_j^2}{n-j} \rightarrow_d \chi^2(p)$ Need exogenous.

When endogenous, $\Phi_{jl} = E(x_t u_{t-j})' E(x_t x_t')^{-1} E(x_t u_{t-l}) / \sigma^2$, $Q_n = n \hat{\rho}' (I - \hat{\Phi})^{-1} \hat{\rho} \rightarrow_d \chi^2(p)$

Correct for AR(1) errors: use FGLS, change Y to $y_t - \rho y_{t-1}$, PW and CO methods. Need strict endogeneity.

Asymptotic variance of OLS estimator: $\text{AVar}(\hat{\beta}_1) = \frac{\text{Var}(\sum r_t u_t)}{(\sum E(r_t^2))^2}$, r_t is residual of x_{t1} on other Xs

Use HAC error(heteroskedasticity and AC consistent); use if serial correlation or strict exogeneity fails.

Ch9. ARIMA

Yule-Walker equations: For AR, calculate ACVF

Ch10. Unit Root

Unit root problems: AR coefs biased towards 0, t-stat not normal, Y and X false correlation.

DF test: $y_t = \alpha + \rho y_{t-1} + e_t$, $H_0 : \rho = 1$, $H_1 : \rho < 1$. T-stat follows DF distribution not normal.

Augmented DF test: $\Delta y_t = \alpha + \theta y_{t-1} + \sum \gamma_i \Delta y_{t-i} + e_t$. Same critical value.(If add trend, different value)

Spurious regression: regress y and x with unit root may lead to high t-stat and R squared.

Cointegration: $y_t = \alpha + \beta x_t + e_t$, $e_t \sim I(0)$ Test :EG-ADF test, perform ADF on regressed residuals.

In cointegration, OLS is consistent even with endogeneity. But in general not normal, need strict endogeneity.

Dynamic OLS estimator: strict exogenous, t-stat asymptotically normal, but serial corr, need HAC se.

Ch12. Matrix OLS

$Y = X\beta + \epsilon$. Assumptions: 1. Linearity 2. Nonsingularity, X full col-rank 3. Strict exogeneity, $E(\epsilon_i | X) = 0$ 4. Spherical error variance $\text{Cov}(\epsilon) = \sigma^2 I$. Estimator: $\hat{\beta} = (X'X)^{-1} X'Y$. BLUE with 1-4

$\sum (Y_i - \bar{Y})^2 = \sum (\hat{Y}_i - \bar{Y})^2 + \sum \hat{\epsilon}_i^2$, $\sum Y_i^2 = \sum \hat{Y}_i^2 + \sum \hat{\epsilon}_i^2$, TSS = ESS + SSR

A5: Normality: $\epsilon | X \sim N(0, \sigma^2 I_n)$ (Implies 3 and 4). if $\epsilon \sim N(0, \Sigma)$, $\epsilon' \Sigma^{-1} \epsilon \sim \chi^2(n)$

T-test: $H_0 : c'\beta = r$, $T_n = \frac{c'\hat{\beta} - r}{\sqrt{\sigma^2 c'(X'X)^{-1} c}}$

F-test: $H_0 : R\beta = r$, $F_n = \frac{1}{q} (R\hat{\beta} - r)' [s^2 R(X'X)^{-1} R']^{-1} (R\hat{\beta} - r) \sim F(q, n - k)$

2SLS: $\hat{\beta}^{2SLS} = (\hat{X}' \hat{X})^{-1} \hat{X}' Y$. When just identified, same as ILS, IV, MOM.